Project 4: Play Chess against AI

Overview

The crucial moment in the history of Artificial Intelligence was when the AI model beat the chess world champion, Garry Kasparov, in 1997. The project will let students create an AI model to play the (simpler and smaller version of) chess. Such learning methods are used very commonly in most AI applications.

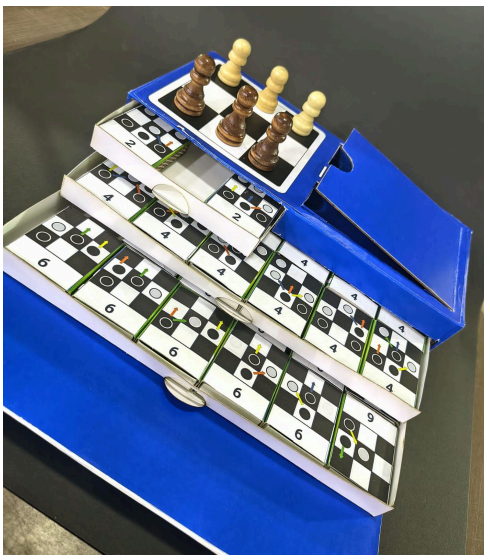
Learning Outcomes

- *Introduction to AI methods.* The game learning requires applying AI methods to train the model.
- *Symmetry and Abstraction. The number of possibilities of the board is reduced with the help of symmetry.*
- *Game Strategy.* Simplicity in the game, yet non-trivial to figure out the winning strategy, provides a platform for an experiential approach in teaching/learning.
- *Visualising Learning Graph.* The graph of the number of plays and winning by the learning AI model provides insight into the training process.

Material Required (for a group of 3/4 students)

- Print of this A4 sheet template
- Matchboxes
- Glue
- Colored beads (4 colours, 20 each) or dal grains with different colours





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Project Plan

Project Activity	Time	Description
Launch the Game	10-15 mins	The teacher will introduce the simpler version of chess with only pawns. Demonstrate the game by playing it with students for a couple of rounds to let them get clarity about the rules. Template: tinyurl.com/chesstemplate Also Teacher can take the assistance of matchbox chess online version. Online : https://tinyurl.com/matchboxchessonline
Play & Explore	20-25 mins	The students will form a group of 3 or 4 and play the game among themselves. The teacher suggests a few explorations while letting students play.
Designing an AI Model	1 week	The students will design a training model to play the game using matchboxes.
Visualisation and Extension	1 week	Visualise the learning curve by plotting the graph: number of games played vs wins. Extend the matchbox based AI model to another game - *Mini Checkers*.

The Game

A simplified version of chess played on a 3 x 3 grid with six pawns (3 white and 3 black).

Rules:

- Pawn can move one square forward into an empty square, or
- A pawn can move diagonally left or right to capture an opponent's pawn, which is removed from the board.

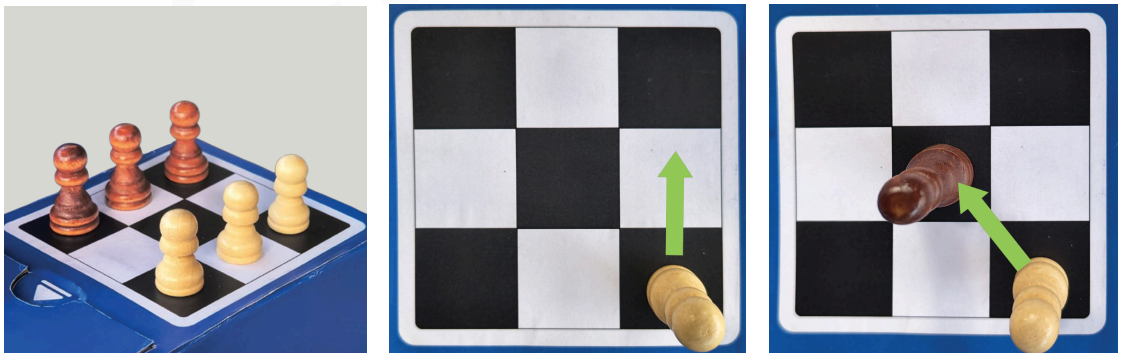


Figure: Setup and Moves of the pawn in the game

Winning Conditions:

- One of their pawns makes it to the opposite side of the board.
- All the opponent's pawns have been captured.
- The opponent has no legal moves to make

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Designing an AI Model

White always starts the game. Turns alternate between white and black, with only one move allowed each turn. Student (player 1) always plays as white. The Matchbox (player 2) always plays as black.

Training the Matchboxes

1. The student (player 1) starts the game and will play a legal move of their choice.
2. On the matchbox's turn, find the matchbox that resembles the current board position (Note: the positions of the pawns may be a mirror image of what is on the matchbox)
3. Shake the matchbox, open the drawer, pick a random bead without looking, and note its colour. The colour bead matches each arrow's colour in the pattern. Which represents the matchbox's move. (Note: Picking an orange bead will result in making the matchbox move with the orange colour arrow.)

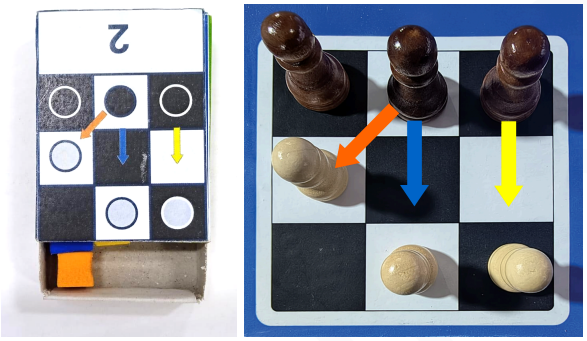


Figure: Bead and Move

4. Players alternate moves, moving one piece at a time. Continue this until one player has either made the opponent unable to make any move, eliminated all of the pawns, or gotten one pawn to the other side of the board.
5. At the end of the game:
    a. If the matchbox wins, "**Reward**" it by putting all the colour beads back inside.
    b. If the matchbox is lost, "**Punish**" it by taking out the bead from its last move.
6. As you play more games against the matchbox based AI model, it will perform better. This process simply teaches the machine what moves caused it to lose, and eliminates those moves, leaving only winning moves. After about 30 games or so, the matchbox's AI model becomes unbeatable. No matter what you move as Player 1, it will always win.

Possible States in the Game

Although many moves seem possible, after removing all illegal moves and symmetrical duplicates (mirror reflections), the total number of possible board states of the game reduces to 19. These 19 states are further divided into 2nd Chance, 4th Chance, and 6th Chance (which are the machine's moves).

- 2nd Chance (Black's First Move)
  In this case, the total number of states is 3, but after removing symmetrical duplicates, it reduces to 2. For each state, we have some possible moves, shown at

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the top of the match box as a coloured arrow, and for each move, we include a bead of the same colour.

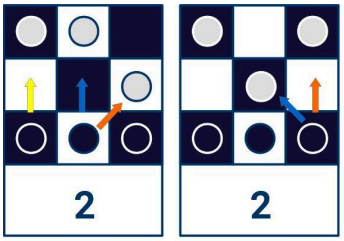


Figure: Possible board states and Machine moves in 2nd Chance.

- 4th Chance (Black's Second Move)
  Similarly, for this, the total number of possible states is 10.

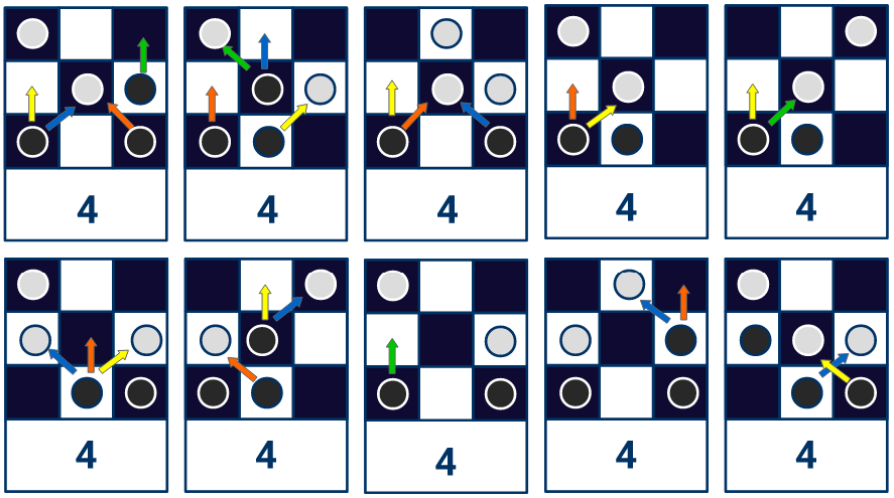


Figure: Possible board states and Machine moves in the 4th Chance.

- 6th Chance (Black's Third Move)
  Similarly, for this, the total number of possible states is 7.

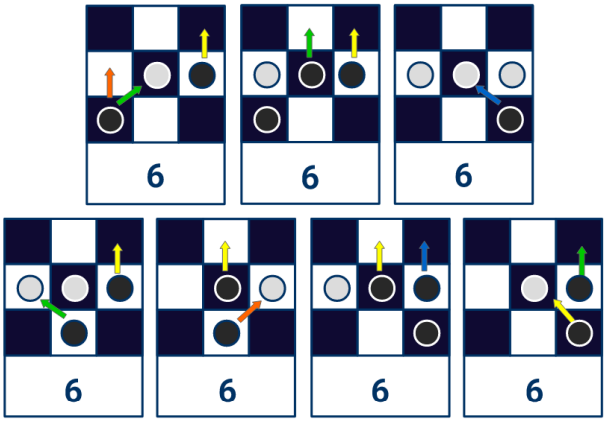


Figure: Possible board states and Machine moves in the 6th Chance.

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Reinforcement-based Machine Learning

The matchbox based AI model learns through experiences, just like us. The model takes actions, receives rewards for good choices (like winning a game), or punishments for bad ones (like losing the game), and gradually improves its strategy to maximise rewards over time. This type of AI learning algorithms are called *Reinforcement-based Machine Learning*. In the matchbox computer, each matchbox acts as the neuron for a specific board state.

Outcome	Action	Effect on Matchbox
Win	Reward	Return all beads to the box (preserve all actions)
Loss	Punishment	Remove the bead matching the final (losing) action (eliminate poor choice)

Early on, the matchbox computer makes moves randomly and often loses. After 30 games or so, poor moves are eliminated, leaving only winning moves.

Explore

1. How could you modify the game to speed up learning ?

    a. Reward winning moves by adding extra beads for those moves
    b. Punish every move in every game, regardless of outcome
    c. Remove the random draw and always pick the most frequent bead
    d. Reset all boxes after each game to encourage fresh exploration

2. Suppose you run two identical matchbox computers against each other, replacing the human player. What outcome do you expect, assuming standard reinforcement rules?

3. Try to play and make a graph of Winning/Losing against the Number of games played. One such example graph is shown below.

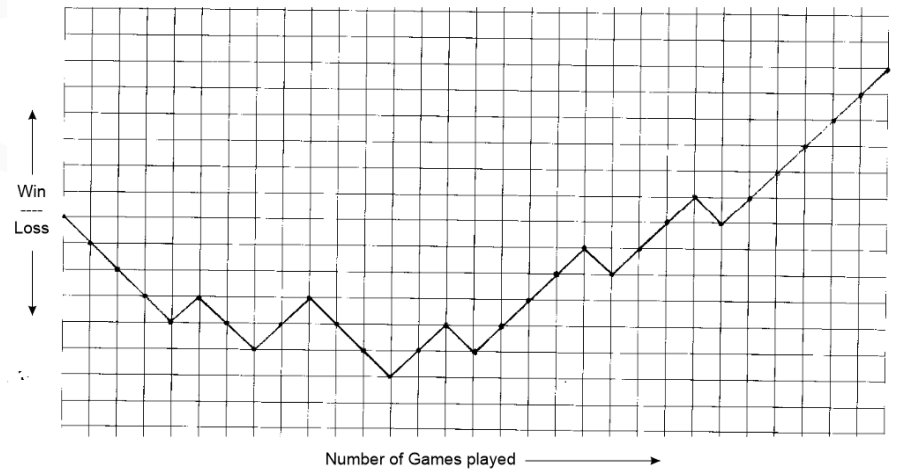


Figure: Learning Curve for the First 30 Games

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Extension: AI Model for Mini-Checkers Game

Mini-Checkers is a simplified version of checkers on a 4x4 board. White (student - player 1) goes first, and the AI model plays as Black.

- *Pieces Movement*: Pieces move diagonally forward to an adjacent, unoccupied dark square. Capturing: A jump involves moving over an opponent's piece to an empty square immediately beyond it.
- *Multiple Jumps*: If a piece captures an opponent's piece and can jump another (or more) in the same turn, it must do so.
- *Kinging*: A piece reaching the farthest row (the last row) becomes a king, allowing it to move forward and backward.
- *Winning*: The game is won by capturing all the opponent's pieces or if the opponent has no valid moves remaining

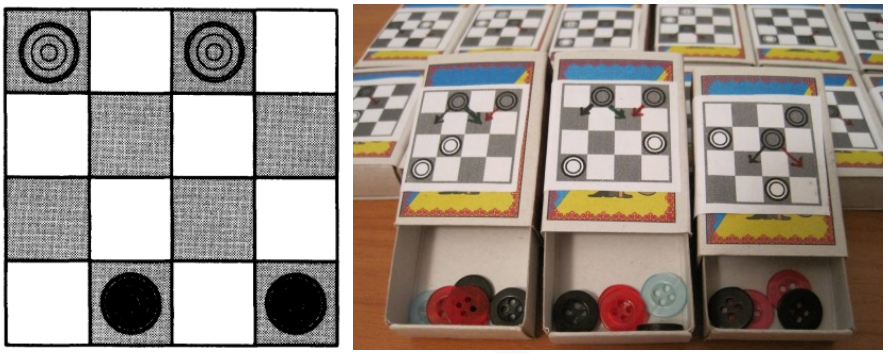


Figure: The setup for the Mini Checkers Game

Refrence: A matchbox game learning-machine by Martin Gardner. Scientific American, March 1962

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